

Minimal and intrinsic topologies on monoids of elementary embeddings

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joint work with J. de la Nuez Gonzalez, Zaniar Ghadernezhad, and Michael Pinsker

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Outline

- ① Preliminaries
- ② The Zariski topology
- ③ Proving minimality

$\text{Aut}(\mathbb{A})$ and $\text{EEmb}(\mathbb{A})$.

$\mathbb{A} :=$ a (first-order) structure.

We are interested in two natural spaces of symmetries:

- $\text{Aut}(\mathbb{A}) :=$ automorphism group of $\mathbb{A}$;
- $\text{EEmb}(\mathbb{A}) :=$ monoid of elementary embeddings of $\mathbb{A}$
($f : \mathbb{A} \rightarrow \mathbb{A}$ preserving the truth of first-order formulas).

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Examples

- $\mathbb{A} := (\mathbb{N}; =)$.
 - $\text{Aut}(\mathbb{A}) := \text{Sym}(\mathbb{N})$;
 - $\text{EEmb}(\mathbb{A}) := \text{Inj}(\mathbb{N})$.
- $\mathbb{A} := (\mathbb{F}_q^{\mathbb{N}_0}; 0; +; (\lambda_r)_{r \in \mathbb{F}_q})$ (countable vector space over $\mathbb{F}_q$).
 - $\text{Aut}(\mathbb{A}) := \text{GL}(\mathbb{N}_0, \mathbb{F}_q)$;
 - $\text{EEmb}(\mathbb{A}) :=$ injective linear transformations $T : \mathbb{F}_q^{\mathbb{N}_0} \rightarrow \mathbb{F}_q^{\mathbb{N}_0}$.

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Topologies on spaces of symmetries

We can endow $S \in \{\text{Aut}(\mathbb{A}), \text{EEEmb}(\mathbb{A})\}$ with the **topology of pointwise convergence** τ_{pw} :

$$\text{for } a, b \in \mathbb{A}, \mathcal{U}_{a,b} := \{f \in S \mid f(a) = b\} ,$$

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When $\mathbb{A}$ is countable, with τ_{pw} :

- $\text{Aut}(\mathbb{A})$ is a Polish* topological group;
- $\text{EEmb}(\mathbb{A})$ is a Polish topological monoid.

***Polish:** separable and completely metrizable.

ω -categorical structures

Definition (ω -categoricity)

$|\Omega| = \aleph_0$. $G \curvearrowright \Omega$ is **oligomorphic** if for all $n \in \mathbb{N}$, $G \curvearrowright \Omega^n$ has finitely many orbits in its diagonal action

$$g \circ (a_1, \dots, a_n) = (ga_1, \dots, ga_n).$$

$\mathbb{A}$ is **ω -categorical** if $\text{Aut}(\mathbb{A})$ is oligomorphic.

Examples

- $(\mathbb{N}; =)$;
- $(\mathbb{Q}; <)$;
- the countable random graph $\mathcal{R}$;
- countably infinite vector spaces over finite fields.

👁 In this context, $\text{EEmb}(\mathbb{A})$ is the τ_{pw} -closure of $\text{Aut}(\mathbb{A})$ in $\mathbb{A}^{\mathbb{A}}$.

- So, from now on, write $G := \text{Aut}(\mathbb{A})$ and $\overline{G} := \text{EEmb}(\mathbb{A})$.

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Why care about topology?

Let $S \in \{\text{Aut}(\mathbb{A}), \text{EEmb}(\mathbb{A})\}$ for $\mathbb{A}$ ω -categorical.

- S with τ_{pw} encodes $\mathbb{A}$ up to bi-interpretability (Ahlbrandt and Ziegler 1986; Lascar 1991); [► More on bi-interpretations](#)
- If we can distinguish τ_{pw} amongst the topologies compatible with the algebraic structure of S , S encodes $\mathbb{A}$ as a group/monoid!
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Definition (Minimality)

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Examples (for which τ_{pw} is minimal on $\text{Aut}(\mathbb{A})$)

- $(\mathbb{N}; =)$ (Gaughan 1967);
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Examples (for which τ_{pw} is **NOT** minimal on $\text{Aut}(\mathbb{A})$)

- $(\mathbb{Q}; <)$ (Chang and Gartside 2017): we can define a coarser **interval topology** τ_{int} with, for $a, b < c \in \mathbb{Q}$:

$$\mathcal{U}_{a,(b,c)} := \{f \in \text{Aut}(\mathbb{Q}) \mid f(a) \in (b, c)\},$$

forming a sub-basis of open sets.

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Examples (for which τ_{pw} is NOT minimal on $\text{Aut}(\mathbb{A})$)

- the **random bipartite graph** $\mathcal{B}$: we can define a coarser **localised topology** τ_{pw}^X (where X is one of the $\text{Aut}(\mathcal{B})$ -orbits) with for $a, b \in X$:

$$\mathcal{U}_{a,b} := \{f \in \text{Aut}(\mathcal{B}) \mid f(a) = b\},$$

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A **semigroup** topology τ on S is **minimal** if there are no strictly coarser Hausdorff **semigroup** topologies.

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Question

Are there any oligomorphic permutation groups for which τ_{pw} is not minimal on $\overline{G}$?[†]

[†]cf. Marimon and Pinsker 2025; Pinsker and Schindler 2023; Elliott, Jonušas, Mesyan, Mitchell, Morayne, and Peresse 2023.

Algebraic closure

Definition (Algebraic closure)

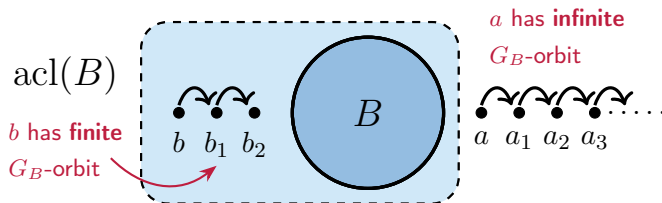
$G \curvearrowright \Omega$. For $B \subseteq \Omega$ finite,

$$\text{acl}(B) := \{b \in \Omega \mid \text{the } G_B\text{-orbit of } b \text{ is finite}\}.$$

For $C \subseteq \Omega$,

$$\text{acl}(C) = \bigcup_{B \subseteq C \text{ finite}} \text{acl}(B).$$

$\mathbb{A}$ has **no algebraicity** if $\text{acl}(B) = B$ always (for $G = \text{Aut}(\mathbb{A})$).



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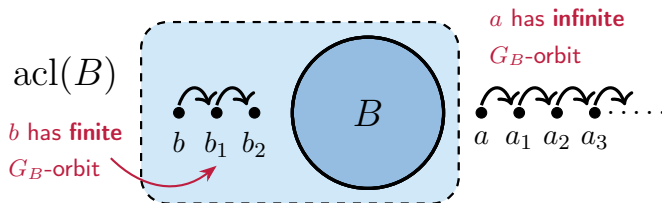
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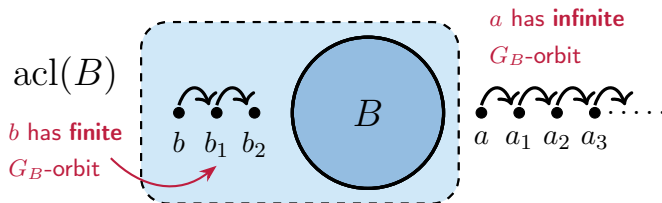
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Examples

- with **no** algebraicity: $(\mathbb{N}; =)$, $(\mathbb{Q}; <)$, $\mathcal{R}$, and $\mathcal{B}$;
- with algebraicity: vector spaces, and $\aleph_0$ disjoint copies of K_2 .

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For $\mathbb{A}$ ω -categorical acl is **locally finite**: for finite B , $\text{acl}(B)$ is finite.

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Theorem (Pinsker and Schindler 2023[†])

Let $\mathbb{A}$ be ω -categorical and with no algebraicity.

Then, τ_{pw} is minimal on $\text{EEmb}(\mathbb{A})$.

There are predecessors in Elliott, Jonušas, Mesyan, Mitchell, Morayne, and Peresse 2023 and Elliott, Jonušas, Mitchell, Peresse, and Pinsker 2023.

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💡 This is very different from the case of $\text{Aut}(\mathbb{A})$!

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Let $\mathbb{A}$ be ω -categorical and with no algebraicity.

Then, τ_{pw} is minimal on $\text{EEmb}(\mathbb{A})$ *because it coincides with the semigroup Zariski topology τ_Z .*

The Zariski topology

Definition (the Zariski topology)

$\mathcal{S} :=$ a semigroup. The (semigroup) **Zariski** topology τ_Z has a sub-basis of open sets given by **solutions to semigroup disequalities**:

$$\{s \in \mathcal{S} \mid t_k s t_{k-1} s \dots t_1 s t_0 \neq q_l s q_{l-1} s \dots q_1 s q_0\},$$

for $k, l \geq 1$ and $t_0, \dots, t_k, q_0, \dots, q_l \in \mathcal{S}$.

τ_Z is contained in any Hausdorff semigroup topology on $\mathcal{S}$.
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When is $\tau_Z \subsetneq \tau_{pw}$?

Pinsker and Schindler 2023 show that for $\mathbb{A} := \aleph_0$ -many copies of K_2 , $\tau_Z \subsetneq \tau_{pw}$ on $\mathbf{EEmb}(\mathbb{A})$ answering a question of Elliott, Jonušas, Mesyan, Mitchell, Morayne, and Peresse 2023.

We prove the following more general statement:

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Theorem (de la Nuez Gonzalez, Ghadernezhad, M., Pinsker 2026)

Let $G \curvearrowright \Omega$ have locally finite algebraic closure and non-trivial centre. Then, the semigroup Zariski topology τ_Z on $\overline{G}$ is not Hausdorff.

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What about minimality of τ_{pw} with algebraicity?

By our theorem, the standard method to prove minimality for τ_{pw} cannot not work in many natural ω -categorical structures with algebraicity:

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Can we still prove minimality by other means?

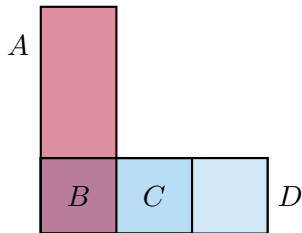
Base monotonicity

Definition (Base monotonicity)

$G \curvearrowright \Omega$ with locally finite acl .

acl satisfies **base monotonicity** if for finite algebraically closed $A, B \subseteq C \subseteq D$,

if $\text{acl}(A \cup B) \cap D = B$, then $\text{acl}(A \cup C) \cap D = C$.



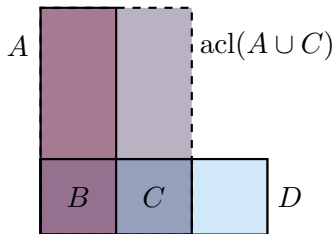
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Morally: acl satisfies base monotonicity when it is “not more complicated than in a vector space”.

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Let $G \curvearrowright \Omega$ be oligomorphic and such that acl satisfies base monotonicity. Then, τ_{pw} is minimal on $\overline{G}$.

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► More on the proof

Thank you!

For $\mathbb{A}$ ω -categorical

- We show τ_Z is not Hausdorff on $\mathbf{EEmb}(\mathbb{A})$ whenever $\mathbf{Aut}(\mathbb{A})$ has non-trivial centre;
- We find new ways to prove minimality of τ_{pw} as long as $\mathbf{acl}$ is sufficiently tame.

Survey:



Paper:



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Bi-interpretations

Definition

An **interpretation** of $\mathbb{B}$ in $\mathbb{A}$ is a partial surjection $I : A^d \rightarrow B$ for some $d \in \mathbb{N}$ such that for each relation R of $\mathbb{B}$ defined by an atomic formula, $I^{-1}(R)$ is definable in $\mathbb{A}$ (without parameters).

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Example

$(\mathbb{C}; 0; 1; \times; +)$ is interpretable in $(\mathbb{R}; 0; 1; \times; +)$ (by the Argand plane).

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Interpretations compose naturally.

If I is an interpretation of $\mathbb{B}$ in $\mathbb{A}$ and J is an interpretation of $\mathbb{A}$ in $\mathbb{B}$, $\mathbb{A}$ and $\mathbb{B}$ are **bi-interpretable** if $I \circ J$ and $J \circ I$ are definable in $\mathbb{B}$ and $\mathbb{A}$ respectively.

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Example

$(\mathbb{N}; =)$ is bi-interpretable with the **Johnson graph** $\mathfrak{J}(2)$:

- Domain: $[\mathbb{N}]^2$ (unordered pairs);
- Edges $E := \{(\{a, b\}, \{c, d\}) \in ([\mathbb{N}]^2)^2 \mid |\{a, b\} \cap \{c, d\}| = 1\}$.

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Theorem (Coquand in Ahlbrandt and Ziegler 1986, cf. Lascar 1991)

Let $\mathbb{A}$ and $\mathbb{B}$ be ω -categorical structures. Then, TFAE:

- $\mathbb{A}$ and $\mathbb{B}$ are bi-interpretable;
- $\text{Aut}(\mathbb{A})$ and $\text{Aut}(\mathbb{B})$ are isomorphic as topological groups;
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Proof sketch

Proof sketch of the theorem.

A semigroup disequality $w(s)$

$$t_k s t_{k-1} s \dots t_1 s t_0 \neq q_l s q_{l-1} s \dots q_1 s q_0$$

is **singular** if $k = l$ and **non-singular** otherwise.

Write $\mathcal{O}_{w(s)}$ for the τ_Z -open set of solutions to $w(s)$.

- if $w(s)$ is singular, for $\gamma \in Z(G)$,
 $1 \in \mathcal{O}_{w(s)}$ if and only if $\gamma \in \mathcal{O}_{w(s)}$;
- if $w(s)$ is non-singular, then $\overline{G} \setminus \mathcal{O}_{w(s)}$ is nowhere dense in $\overline{G}$
 (wrt τ_{pw}) (use Neumann's Lemma).
- Because of this any $\gamma \in Z(G)$ cannot be separated from 1 by
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Lovely pairs

See Berenstein and Vassiliev 2010 (cf. Ben-Yaacov, Pillay, and Vassiliev 2003):

Definition (Lovely pair)

$\mathbb{A}$ ω -categorical. $L \subseteq \mathbb{A}$ algebraically closed is a **lovely pair** if

- (coheir) for finite algebraically closed A, B with $\text{acl}(B) \cap \text{acl}((L \cap B) \cup A) = \text{acl}(L \cap B)$, there is $g \in G_B$ such that $g(A) \subseteq L$;
- (extension) for finite algebraically closed A, B there is $g \in G_B$ with $\text{acl}(g(A) \cup B) \cap L = B$.

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Examples

- in $(\mathbb{N}; =)$ this is an infinite-coinfinite set;
- in $(\mathbb{Q}; <)$ this is a dense codense set;
- in vector space $\mathbb{F}_q^{\aleph_0}$ this is an infinite-dimensional subspace of infinite codimension.

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We need a slight variant of a lovely pair (but in all examples this is just a lovely pair);

What is left out?

- Droste's almost homogeneous semilinear orders
(but for these we can prove $\tau_{\text{pw}} = \tau_Z$);
- Hrushovski constructions
(but these seem to be amenable to a variant of our proof);
- atomless Boolean algebras
(τ_{pw} should be minimal by other methods).

Rigidity and dispersivity

$G \curvearrowright \Omega$. Consider a semigroup topology $\tau \subseteq \tau_{\text{pw}}$ on $\overline{G}$.

Definitions (Rigidity and dispersivity)

$t \in \overline{G}$ is **τ -rigid** on $a \in \Omega$ if there is some τ -neighbourhood $\mathcal{V}$ of t such that, for all $s \in \mathcal{V}$, $s(a) \in \text{acl}(t(a))$.

$t \in \overline{G}$ is **τ -dispersive** on $a \in \Omega$ if for every finite $B \subseteq \Omega$ and any τ -neighbourhood $\mathcal{V}$ of t there is $s \in \mathcal{V}$ such that $a \notin s(B)$.

 $t \in \overline{G}$ is τ_{pw} -rigid on Ω and not τ_{pw} -dispersive on $\text{Im}(t)$.

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Lemma (de la Nuez Gonzalez, Ghadernezhad, M., and Pinsker 2026)

$G \curvearrowright \Omega$ with locally finite acl . Suppose that $\tau \subsetneq \tau_{\text{pw}}$ is a semigroup topology and every $t \in \overline{G}$ is τ -rigid on Ω . Then, τ is not T_1 .

Proof idea of the main theorem

Proof idea.

Suppose $\tau \subsetneq \tau_{\text{pw}}$ is a Hausdorff semigroup topology. Consider

$$\Delta := \{a \in \Omega \mid \text{every } s \in \overline{G} \text{ is } \tau\text{-rigid on } a\}.$$

By previous Lemma, $\Delta \subsetneq \Omega$.

We can find $t \in \overline{G}$ satisfying a strong form of τ -dispersivity on $\Omega \setminus \Delta$.

By **base monotonicity**, find a “generic” copy $\mathbb{A}^*$ of $\mathbb{A}$ within itself.

In particular,

$$\mathcal{U} := \{s \in \overline{G} \mid \text{Im}(s) \subsetneq \mathbb{A}^*\}$$

is τ_Z -open (and so τ -open) and a neighbourhood of t .

We exploit τ -dispersivity and genericity of $\mathbb{A}^*$ to find:

$u \in \mathcal{U}$ such that $\text{Im}(u) \subseteq \mathbb{A}^*$. (CONTRADICTION!)



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